

YIELDSHIELD: A PREDICTIVE CROP YIELD INTELLIGENCE SYSTEM FOR MUNICIPAL AGRICULTURAL MANAGEMENT

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In Partial Fulfillment of the Requirements for the Course
Capstone Project 1

Bachelor of Science in Information Technology
Major in Data Analytics

by

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In partial fulfillment of the requirements for the course Capstone Project 1 of the degree Bachelor of Science in Information Technology major in Data Analytics, this capstone project proposal entitled **“YIELDSHIELD: A PREDICTIVE CROP YIELD INTELLIGENCE SYSTEM FOR MUNICIPAL AGRICULTURAL MANAGEMENT”**, which was prepared and submitted by LENDYL TYRON L. ESTABAYA, DAVE JERICHO T. DAMIAN, JUSTIN AARON P. FIESTA, ASHLEY N. GAPASIN, ROMNICK JAY O. SUBIDO, is hereby recommended for approval and acceptance.

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GLOSSARY OF TERMS

Analytics Dashboard. A system interface that visually presents crop yield predictions, environmental data insights, and analytical results in the YieldShield platform.

Climate Data. Environmental information such as temperature, rainfall, and humidity used as input variables for crop yield prediction in the system.

CRISP-DM (Cross-Industry Standard Process for Data Mining). A data mining methodology used in the study to guide the processes of data collection, preparation, modeling, and evaluation.

Crop Recommendation. A system-generated suggestion of suitable crops based on analyzed environmental conditions and land characteristics.

Crop Yield. The amount of agricultural produce harvested per unit of land, which serves as the primary output predicted by the system.

Data-Driven Farming. An approach to agricultural decision-making that uses data analysis, predictive models, and environmental information to guide crop selection, planning, and resource management instead of relying solely on traditional practices.

Data Preprocessing. The process of cleaning, organizing, and transforming raw agricultural and environmental data into a format suitable for model development.

Decision Support System (DSS). A component of YieldShield that provides data-driven insights to assist users in making informed agricultural decisions.

Environmental Variables. Factors such as rainfall, temperature, humidity, and soil conditions that influence crop growth and are used as inputs in the predictive model.

Machine Learning. A computational approach used in the system to analyze patterns in data and generate crop yield predictions.

Predictive Model. The core component of the system that processes input data to generate crop yield forecasts using machine learning techniques.

Random Forest Algorithm. A machine learning technique used in the study to improve prediction accuracy by combining multiple decision trees.

Rapid Application Development (RAD). A system development methodology used to design and develop the YieldShield system through iterative prototyping and user feedback.

System Usability Scale (SUS). A standardized tool used to evaluate the usability, efficiency, and user satisfaction of the system.

User Interface (UI). The visual layout of the system that allows users to interact with features such as data input, dashboard, and reports.

Yield Prediction. The process of estimating future crop production based on environmental and historical data using the system.

DECLARATION OF ORIGINALITY

We, the undersigned proponents, hereby declare that this manuscript, titled **“YIELDSHIELD: A PREDICTIVE CROP YIELD INTELLIGENCE SYSTEM FOR MUNICIPAL AGRICULTURAL MANAGEMENT”**, is a true reflection of our own research and efforts. We further certify that:

- a) all data and materials incorporated herein are original, except where explicitly cited and acknowledged from other authoritative sources;
- b) standard rules for proper citation and referencing have been strictly observed;
- c) the study was conducted in full compliance with established ethical guidelines and institutional standards of academic integrity; and
- d) this project proposal has not been previously submitted, in whole or in part, to any other higher education institution for an academic degree, nor has it been presented in any local or international forums, workshops, or competitions prior to this submission.

Project Proponents:

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DECLARATION OF AI-ASSISTED TOOLS USAGE

We, the undersigned proponents of the project entitled **“YIELDSHIELD: A Predictive Crop Yield Intelligence System for Municipal Agricultural Management”** hereby declare the extent and nature of the Artificial Intelligence (AI)-assisted tools utilized in the completion of this Capstone Project Proposal.

We acknowledge that AI-assisted tools were used solely as supplementary support in the development of this project, particularly in *documentation and writing support*, such as aiding in refining grammar, clarity, organization, and overall coherence of the manuscript.

We affirm that all AI-assisted outputs have been thoroughly reviewed, validated, and significantly revised by the proponents. The final outputs of this project reflect our original work, critical analysis, and academic responsibility. We fully understand that the use of AI tools does not substitute for academic integrity; therefore, we assume full responsibility and accountability for the accuracy, originality, and authenticity of this Capstone Project Proposal.

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CHAPTER 1

INTRODUCTION

Project Context

Agriculture plays a critical role in sustaining global food security and economic stability, supporting billions of people worldwide through food production, employment, and raw material supply, as reported by the Food and Agriculture Organization. Furthermore, agriculture employs more than 26% of the global workforce and remains the primary livelihood source for rural populations (FAO, 2023). In addition, the sector contributes significantly to national economies, particularly in developing countries where food production and rural development are closely interconnected (World Bank, 2023). However, despite its importance, agriculture is increasingly threatened by challenges such as climate change, rapid population growth, and environmental degradation, all of which directly affect its productivity and long-term sustainability (Intergovernmental Panel on Climate Change, 2022).

These challenges are further intensified by the growing unpredictability of weather patterns, declining soil fertility, and the continuous rise in global food demand (FAO, 2023). Global food demand is projected to increase by approximately 50% by 2050, placing additional pressure on agricultural systems (Food and Agriculture Organization, 2023). Consequently, farmers are placed under pressure to produce more food using limited and often degrading resources (World Bank, 2023). This situation makes it difficult to maintain stable crop yields and ensure efficient resource utilization (IPCC, 2022). Although governments and agricultural institutions provide support through programs such as seed

distribution, fertilizer subsidies, and technical assistance, many of these initiatives still rely on traditional methods (FAO, 2023; World Bank, 2023). These traditional methods include manual record-keeping, which involves the use of handwritten logs to document farm activities and yields; historical planting practices, where crop decisions are guided by past seasonal patterns rather than current environmental conditions; and experience-based decision-making, where farmers depend on personal or inherited knowledge instead of data-driven analysis or predictive modeling (FAO, 2023). As a result, the lack of integration of data-driven technologies limits the ability of farmers to make accurate, timely, and evidence-based decisions (World Bank, 2023).

In the context of the Philippines, agriculture remains a vital sector contributing approximately 9.4% to the country's Gross Domestic Product (GDP) according to the Philippine Statistics Authority (PSA, 2023). The sector also employs about 22–25% of the labor force, highlighting its importance in rural livelihoods (PSA, 2023). However, the country's geographic location makes it highly susceptible to climate-related risks (IPCC, 2022). The Philippines is ranked among the most climate-vulnerable countries in the world (World Risk Index, 2023) as it experiences an average of 20 typhoons annually (PAGASA, 2023).

Given these challenges, agricultural activities in the Philippines are frequently disrupted due to typhoons and extreme weather conditions (PAGASA, 2023), leading to inconsistent production and crop damage (Department of Agriculture, 2023), which in turn contributes to financial losses for farmers. The value of agricultural production, measured at constant 2018 price levels using 2018 as the base year to remove the effects of inflation and ensure a fair comparison across years, declined from approximately ₱1.76 trillion in

2022 to ₱1.72 trillion in 2023. This reflects a decrease of about ₱40 billion in real agricultural output, or an estimated 2.4% reduction in overall production (PSA, 2023). In addition, agricultural damages caused by typhoons in 2023 reached around ₱13.7 billion, with rice and corn among the most severely affected crops (DA, 2023), making it difficult to maintain stable yields and sustainable livelihoods.

At the local level, particularly in Pangasinan, agriculture is a major economic activity that supports the livelihoods of a large portion of the population. The province is one of the leading agricultural producers in the Ilocos Region, with rice and corn as its primary crops. These agricultural activities contribute significantly to local food supply and income generation for farming communities (PSA, 2023). Despite its importance, agricultural production in Pangasinan is highly affected by environmental and climatic conditions. The province experiences variations in rainfall patterns, temperature fluctuations, and exposure to typhoons, which directly impact crop growth and yield. Northern Luzon, including Pangasinan, frequently experiences extreme weather events (PAGASA, 2023) that disrupt farming activities. In addition, many farmers in the province continue to rely on traditional farming practices based on manual labor, experiential knowledge, and inherited techniques rather than modern mechanized or data-driven systems (Azarias et al., 2023).

Within Pangasinan, municipal-level interventions have been implemented to improve agricultural productivity. In Malasiqui, the Provincial Corporate Farming Program resulted in a 13.6% increase in rice yield, from 4.55 to 5.17 metric tons per hectare (Philippine News Agency, 2025). This improvement demonstrates that localized government initiatives can enhance productivity despite persistent challenges in

conventional farming practices. However, such interventions are still largely program-based and do not fully integrate data-driven tools that can provide continuous, real-time decision support for farmers.

In response to these gaps, this project proposes YieldShield, a web-based system designed to predict crop yield and recommend suitable crops based on environmental and land data. Specifically, the system integrates relevant datasets to generate data-driven farming insights that can guide farmers and agricultural stakeholders in making informed decisions. By doing so, the proposed system aims to enhance productivity, optimize resource utilization, and reduce the risks associated with climate variability. Ultimately, the system promotes more sustainable and resilient agricultural practices in the Philippines.

Purpose and Description of the Project

The purpose of this project is to develop a system that enhances agricultural decision-making by providing data-driven insights for crop planning and yield forecasting. Specifically, the system aims to assist users in analyzing environmental and land-related factors to improve crop selection, optimize resource utilization, and minimize the risks associated with climate variability. By leveraging predictive analytics, the project promotes more efficient, informed, and sustainable farming practices.

The system is designed for two main types of users: administrators and farmers. Administrators manage and maintain the system's agricultural data and monitor overall system usage, ensuring that accurate and up-to-date information is available. Farmers, as the primary end-users, interact with the platform to input their farm data and receive crop yield predictions and recommendations. Through this, the system benefits administrators

by improving data organization and monitoring, while helping farmers make more informed and efficient farming decisions.

YieldShield is a web-based predictive system that integrates agricultural datasets and environmental variables to generate crop yield forecasts and recommendations. The system features a user-friendly dashboard that presents crop yield predictions and relevant insights in a clear and visual format, allowing users to easily interpret results. It also includes a crop recommendation module that suggests suitable crops based on environmental conditions and land characteristics, helping farmers make informed planting decisions. Additionally, the system provides data input modules that enable users to enter key variables such as rainfall, temperature, soil type, and elevation, ensuring that predictions are based on accurate and updated information. A historical data analysis component is incorporated to utilize past records and improve the accuracy of predictions over time. Furthermore, the system includes a report generation feature that produces summarized outputs, supporting effective planning and data-driven decision-making.

The system will incorporate a login and profile configuration feature, ensuring that farmers can access personalized dashboards and view system-generated crop recommendations. Farmers then initiate a farming cycle by entering essential agricultural data into the system, including the number of seedlings planted, crop type, land area, soil conditions, and other relevant environmental and farm-related factors. These inputs are processed by the predictive model, which utilizes historical and environmental datasets to generate crop yield predictions and suggest appropriate farming steps, all of which are displayed on the dashboard. Farmers can also continuously log their farming procedures for monitoring and improved accuracy. Meanwhile, administrators log in to manage and

update the system by feeding newer datasets to enhance model performance, monitoring yield predictions, and reviewing aggregated farmer data across farming cycles. They also handle support requests, such as insurance and seedling assistance, provide necessary interventions, and generate reports for further analysis and decision-making.

To address the limitations of traditional, experience-based farming practices, this project integrates predictive analytics into a localized and user-friendly web platform for agricultural use. It enables farmers to make more informed decisions through accurate and reliable data-driven insights rather than relying solely on intuition or past experience. The system also prioritizes accessibility by converting complex datasets into clear, actionable recommendations, making advanced agricultural intelligence understandable and usable even for users with limited technical expertise.

Objectives of the Project

The main objective of this project is to develop YieldShield, a predictive crop yield intelligence system that utilizes agricultural and environmental data to support crop yield forecasting and agricultural decision-making.

Specifically, it aims to achieve the following specific objectives:

1. To collect and pre-process relevant agricultural and environmental data, from the DA and other sources, for crop yield analysis, such as:
 - a. rainfall amount,
 - b. climate data,
 - c. soil conditions,
 - d. terrain type, and

- e. land size ;
- 2. To develop a crop yield prediction models through various machine learning techniques using the Cross - Industry Standard Process for Data Mining (CRISP-DM) and Rapid Application Development (RAD) Hybrid Methodology ;
- 3. To assess the performance of the developed models using Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and Coefficient of Determination (R^2), in order to determine their predictive accuracy and identify the most suitable model for crop yield prediction.
- 4. To develop a decision support system using Hybrid RAD and CRISP-DM Methodology that integrates the predictive model and provides crop yield forecasts through a user-friendly dashboard; and,
- 5. To evaluate the usability of the developed system using the System Usability Scale (SUS).

Scope and Limitations of the Project

This study focuses on the development of YieldShield, a web-based predictive crop yield intelligence system intended for use in Binalonan, Pangasinan. The system utilizes environmental data, including climate-related variables such as temperature, rainfall, humidity, and real-time weather conditions, together with agricultural data such as crop types, historical yield records, soil characteristics, land area, and elevation. In addition, it integrates real-time weather updates and short-term forecasts to enhance the accuracy of crop yield predictions and crop suitability recommendations. The system also includes an interactive map and geospatial heatmaps for administrators and the Department of

Agriculture, as well as notifications and alerts to indicate potential crop yield risks. Furthermore, the study covers data collection and preprocessing, predictive model development, and the design of a web-based dashboard for visualization and user interaction.

However, the study is subject to several limitations. The system does not incorporate IoT-based sensors or satellite-based monitoring technologies, which may limit the precision of localized data. Instead, weather data will be sourced from a weather data API, ensuring access to regularly updated and location-based environmental information. Moreover, it is restricted to selected crop varieties, specifically palay (rice) and corn, and is geographically limited to Binalonan, Pangasinan. Finally, the accuracy and reliability of the system depend on the quality, completeness, and availability of both historical and real-time data, which may affect the performance of its predictions and recommendations.

CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDIES

Data Collection and Pre-processing

The effectiveness of predictive agricultural systems largely depends on the quality, diversity, and reliability of datasets used in model development. In this study, agricultural data are obtained primarily from existing records and structured datasets provided by the Department of Agriculture and other relevant sources. These include historical crop yield records, soil characteristics, land information, and environmental data such as rainfall, temperature, and humidity. Using these consolidated data sources allows for a comprehensive representation of agricultural conditions suitable for crop yield prediction. (Adeyemi et al., 2023; Rahman et al., 2022; Kumar & Singh, 2024; Chen et al., 2023; Garcia & Lopez, 2022).

Environmental variables such as rainfall, temperature, and soil characteristics are consistently identified in agricultural literature as key determinants of crop yield. Rainfall plays a crucial role in regulating water availability for plant growth, while temperature directly affects crop development stages and physiological processes. Soil characteristics, including nutrient content and moisture retention capacity, significantly influence soil fertility and overall agricultural productivity. Garcia and Lopez (2022) emphasized that climatic variations such as rainfall and temperature directly affect plant growth cycles, while Nguyen et al. (2022) highlighted that soil composition and moisture levels are critical factors in determining crop performance. These variables are commonly treated as

independent predictors in crop yield studies and are essential components in developing accurate predictive models for agricultural systems.

Combining datasets from different domains has also been found to improve predictive performance. Chen et al. (2023) demonstrated that integrating climate and soil datasets produces more robust and accurate models, as it captures both atmospheric and ground-level influences. Similarly, Kim et al. (2022) noted that incorporating historical climate data improves model stability, while Torres et al. (2025) emphasized the importance of historical datasets in identifying long-term trends and seasonal patterns.

In agricultural research, open-source and government datasets are widely used as primary sources of information for predictive modeling and decision-support systems. These datasets provide structured and reliable records of environmental and agricultural variables that can be used for analysis without the need for real-time hardware-based data collection. In this study, data acquisition relies entirely on existing agricultural records and environmental datasets, which are processed and integrated into a web-based dashboard system for visualization, analysis, and crop yield prediction. Additionally, government and open-source agricultural data remain valuable resources as they ensure accessibility, consistency, and relevance for developing data-driven agricultural applications (Adeyemi et al., 2023).

However, raw datasets often contain inconsistencies, missing values, duplicated records, and noise, which can negatively affect model performance and lead to inaccurate predictions if not properly addressed. In agricultural datasets, these issues commonly arise due to variations in data collection methods, incomplete historical records, and human encoding errors. As a result, proper data preprocessing becomes a critical step in ensuring

the reliability and usability of the dataset for predictive modeling. Data preparation techniques such as data cleaning, normalization, transformation, and handling missing values are essential to standardize inputs, reduce bias, and improve overall data quality (Rahman et al., 2022; Alvarez & Cruz, 2024). Data cleaning involves removing or correcting erroneous and inconsistent entries, while normalization ensures that numerical values are scaled within a consistent range to prevent model bias toward larger values. Additionally, data transformation helps convert raw data into suitable formats for analysis, and missing value handling ensures that incomplete records do not distort model training results. Furthermore, structured and well-organized datasets significantly improve machine learning performance by enhancing model stability, training efficiency, and predictive accuracy, as models are better able to identify meaningful patterns within consistent data distributions (Kumar & Singh, 2024).

The reviewed studies emphasize that high-quality and well-prepared datasets are essential in developing reliable agricultural prediction systems. Environmental variables such as rainfall, temperature, soil characteristics, and historical crop yield records were consistently identified as major factors affecting crop productivity. The literature also highlights the importance of data preprocessing techniques, including data cleaning, normalization, transformation, and handling missing values, to improve data quality and predictive accuracy. These findings support the use of structured and integrated datasets in the development of YieldShield. Proper data management also helps improve the consistency and reliability of predictive results.

Predictive Modeling Techniques

Predictive modeling has become a fundamental component of modern agricultural systems, enabling accurate forecasting and data-driven decision-making. Traditional statistical approaches, such as regression analysis, are widely used due to their interpretability and simplicity. However, these methods often struggle to capture nonlinear relationships and complex interactions among variables, which are commonly present in agricultural datasets. Crop yield is influenced by multiple interdependent factors such as rainfall, temperature, soil conditions, and crop type, and these relationships are not always linear or independent. As a result, traditional models may fail to fully represent environmental variability, leading to reduced predictive accuracy (Verma et al., 2022; Singh & Kaur, 2023).

To address these limitations, machine learning and deep learning approaches have been increasingly adopted in agricultural prediction systems. These models are capable of identifying hidden patterns within large and complex datasets, significantly improving prediction performance and reliability (Liakos et al., 2021; Abu Javed & Murad, 2024). Among these techniques, Random Forest is widely recognized for its robustness, resistance to overfitting, and ability to handle high-dimensional data (Jeong et al., 2021; Hassan et al., 2024). Additionally, other models such as Support Vector Machines and neural networks have demonstrated strong predictive capabilities in crop yield estimation (Ahmed et al., 2021; Chowdhury et al., 2023). Ensemble learning techniques further enhance prediction accuracy by combining multiple models to capture complex nonlinear relationships more effectively (Lopez et al., 2024; Wang et al., 2024).

To ensure that predictive models are developed in a structured and systematic manner, the Cross-Industry Standard Process for Data Mining (CRISP-DM) is widely used as a guiding framework. CRISP-DM consists of six iterative phases: business understanding, data understanding, data preparation, modeling, evaluation, and deployment (Wirth & Hipp, 2000; Shearer, 2000). This framework provides a comprehensive and standardized approach to data mining, ensuring that predictive models are both reliable and aligned with real-world objectives.

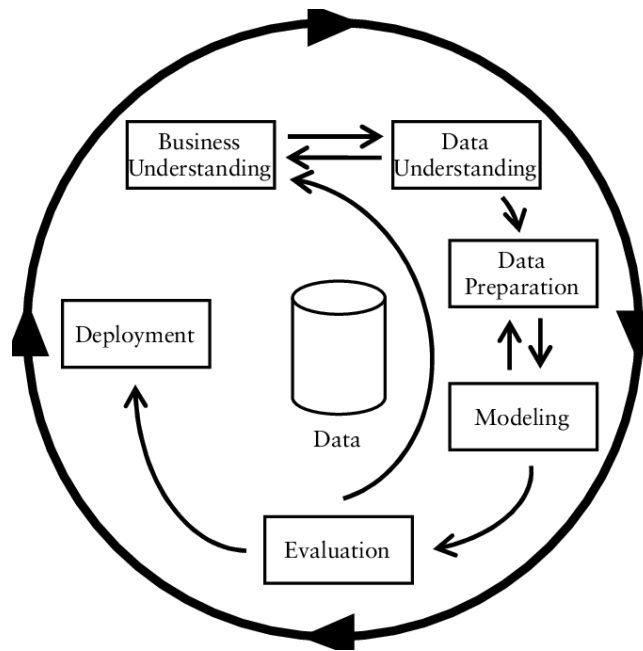


Figure 1: CRISP-DM Model

In the context of agricultural prediction, the business understanding phase involves defining the primary objective of generating accurate crop yield forecasts and actionable recommendations to support decision-making. This is followed by the data understanding phase, where datasets such as rainfall, temperature, soil characteristics, and historical crop yield records are explored to identify patterns, trends, and inconsistencies (Chen et al., 2023; Kim et al., 2022).

The data preparation phase plays a critical role in improving model performance, as agricultural datasets often contain missing values, noise, and inconsistencies. Techniques such as data cleaning, normalization, and transformation are applied to ensure that the data is suitable for analysis (Rahman et al., 2022; Alvarez & Cruz, 2024). Proper data preparation enhances model stability and allows machine learning algorithms to effectively learn meaningful patterns from the data (Kumar & Singh, 2024).

The modeling phase involves applying machine learning algorithms, such as Random Forest and other advanced techniques, to analyze the relationships between environmental variables and crop yield. Different model configurations may be tested to identify the most effective approach for prediction (Liakos et al., 2018; Dela Cruz & Santos, 2025).

One of the key strengths of CRISP-DM is its iterative nature, which allows continuous refinement of the predictive model. If performance is not satisfactory, earlier phases can be revisited to improve results. This adaptability is particularly important in agriculture, where environmental conditions are dynamic and continuously evolving (Abu Javed & Murad, 2024).

The literature shows that predictive modeling is essential in improving agricultural forecasting and decision-making. While traditional statistical methods remain useful, machine learning techniques such as Random Forest, Support Vector Machines, and neural networks were found to provide better accuracy in handling complex agricultural data. The reviewed studies also emphasized the importance of the CRISP-DM framework in guiding systematic model development and continuous improvement. These findings support the implementation of machine learning-based predictive models in YieldShield. The

integration of advanced predictive techniques can also enhance the adaptability of agricultural systems to changing environmental conditions.

Model Performance Evaluation

Model performance evaluation is a critical step in predictive modeling, as it determines how accurately a model can predict outcomes based on input data. In agricultural prediction systems, evaluating model performance is essential to ensure that crop yield forecasts and recommendations are reliable and applicable in real-world scenarios. Since agricultural data is often influenced by multiple dynamic factors such as climate variability, soil conditions, and farming practices, robust evaluation methods are necessary to assess how well a model generalizes across different conditions. Proper evaluation also enables researchers to compare multiple models and identify the most suitable approach for a specific agricultural application (Hassan et al., 2024; Wang et al., 2024).

Among the most commonly used evaluation metrics are Mean Absolute Error (MAE) and Root Mean Square Error (RMSE), which quantify the difference between predicted and actual values. MAE calculates the average magnitude of errors, providing a straightforward and interpretable measure of prediction accuracy. In contrast, RMSE assigns greater weight to larger errors, making it more sensitive to outliers and extreme prediction deviations. These characteristics make both metrics complementary, as MAE reflects overall model consistency while RMSE highlights the presence of significant prediction errors. Recent agricultural studies frequently use both metrics together to obtain

a more balanced and comprehensive evaluation of model performance (Abu Jaded & Murad, 2024; Dela Cruz & Santos, 2025).

In addition to error-based metrics, statistical measures such as the coefficient of determination (R^2) are widely used to evaluate predictive models. R^2 indicates the proportion of variance in the observed data that is explained by the model, providing insight into how well the model captures underlying data patterns. A higher R^2 value suggests stronger predictive capability and better model fit. However, relying solely on R^2 can be misleading, as it does not account for prediction errors directly. Therefore, combining R^2 with MAE and RMSE provides a more complete understanding of model performance (Lopez et al., 2024; Hassan et al., 2024).

Recent literature also emphasizes that the selection of evaluation metrics should be aligned with the specific objectives of the study and the characteristics of the dataset. In agricultural applications where large prediction errors can significantly impact decision-making, such as crop yield estimation or resource allocation, RMSE is often preferred due to its sensitivity to extreme values. On the other hand, MAE is more suitable in scenarios where a consistent and balanced measure of error is required across all predictions. This highlights the importance of using multiple evaluation metrics to avoid biased or incomplete assessments of model performance (Wang et al., 2024).

Furthermore, model validation techniques play a crucial role in ensuring that predictive models perform well on unseen data. One commonly used approach is cross-validation, where the dataset is divided into multiple subsets, and the model is trained and tested on different combinations of these subsets. This method helps reduce overfitting and provides a more reliable estimate of how the model will perform in real-world applications.

By validating models across different data splits, researchers can ensure that the model is not overly dependent on a specific dataset and can generalize effectively to new conditions (Abu Javed & Murad, 2024; Hassan et al., 2024).

In addition, recent studies highlight the importance of continuous model evaluation and updating, particularly in agriculture where environmental conditions and data patterns change over time. Models that are regularly evaluated and retrained using updated datasets tend to maintain higher accuracy and reliability compared to static models. This ongoing evaluation process ensures that predictive systems remain relevant and capable of supporting timely and data-driven decision-making (Wang et al., 2024; Dela Cruz & Santos, 2025).

The reviewed studies highlight the importance of evaluating predictive models to ensure accuracy and reliability in agricultural systems. Metrics such as MAE, RMSE, and R^2 were commonly used to measure prediction performance and model effectiveness. Researchers also emphasized the value of cross-validation in reducing overfitting and improving model generalization. Overall, the literature supports the use of multiple evaluation methods to ensure dependable crop yield predictions in YieldShield. Continuous model evaluation also helps maintain system effectiveness as agricultural data and environmental conditions evolve over time.

Development of Decision Support Systems

Decision Support Systems (DSS) play a critical role in transforming raw agricultural data into actionable insights that support complex and time-sensitive decision-making processes. These systems integrate databases, analytical models, and user-friendly

interfaces to assist users in evaluating different scenarios and selecting optimal strategies. In agriculture, DSS are particularly valuable due to the sector's dependence on dynamic and interrelated variables such as climate conditions, soil characteristics, crop types, and resource availability (Khan et al., 2021; Zhai et al., 2021).

Modern agricultural Decision Support Systems (DSS) increasingly incorporate predictive analytics, simulation techniques, and real-time data integration to enhance decision-making capabilities. These systems utilize data-driven approaches to support crop planning, yield forecasting, and farm management by analyzing environmental and agricultural variables. Recent studies highlight that integrating machine learning models and real-time data sources significantly improves the accuracy and responsiveness of agricultural decision support systems, enabling more efficient and adaptive farm management practices (Alharbi et al., 2024; Reyes et al., 2023). Furthermore, simulation and data-driven modeling approaches allow users to evaluate different agricultural scenarios and assess the potential impact of varying environmental conditions on crop performance, leading to more informed and strategic decision-making (Wang et al., 2024; Abu Jaded & Murad, 2024).

Visualization is another essential component of DSS, as it improves the interpretability of complex analytical outputs. Dashboard-based systems present data through charts, graphs, and interactive elements, allowing users to quickly identify trends and patterns. This is particularly important for farmers and stakeholders with limited technical expertise, as it simplifies data interpretation and enhances decision-making efficiency (Reyes et al., 2023; Perez et al., 2022).

To effectively develop DSS, an appropriate system development methodology must be applied. Rapid Application Development (RAD) is widely used due to its emphasis on iterative design, rapid prototyping, and continuous user feedback. Unlike traditional development models, RAD focuses on quickly building functional system components and refining them through repeated cycles based on user interaction. This approach enables faster system development while ensuring that the final system aligns with user needs and practical agricultural applications (Reyes et al., 2023; Alharbi et al., 2024).

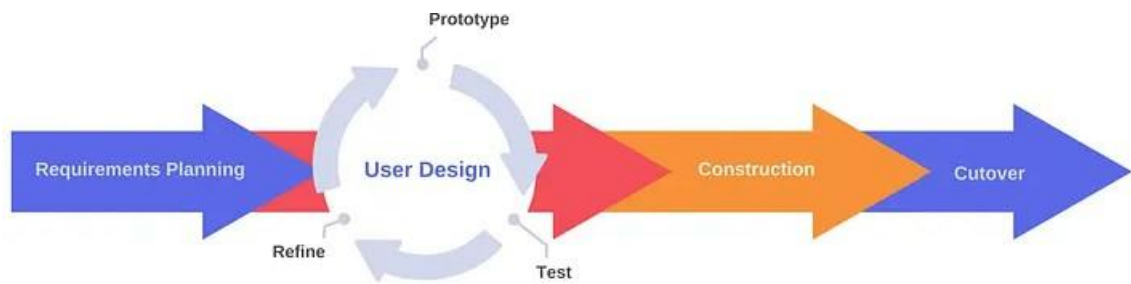


Figure 2: Rapid Application Development Model

Rapid Application Development (RAD) enables developers to rapidly create prototypes of DSS components, such as dashboards, data input modules, and visualization tools. These prototypes are continuously tested and refined through iterative development cycles, allowing developers to detect design issues, improve functionality, and enhance usability at an early stage. This iterative prototyping approach reduces development time while ensuring that system features are aligned with user requirements and real-world agricultural applications. In decision support systems, rapid prototyping is particularly valuable because it allows stakeholders to interact with early versions of the system and provide feedback that directly influences system improvements. Studies have shown that incorporating iterative design and user feedback significantly enhances system usability, efficiency, and overall effectiveness in agricultural and web-based DSS environments

(Reyes et al., 2023; Alharbi et al., 2024; Omar et al., 2021). By continuously refining system components based on user interaction, RAD supports the development of flexible, user-centered systems that can adapt to changing requirements and deliver more practical decision-making support.

One of the key advantages of RAD is its ability to incorporate user feedback throughout the development process. Farmers, agricultural officers, and other stakeholders can evaluate system features and provide insights on usability, clarity, and functionality. This feedback is then used to refine the system, ensuring that it aligns with real-world agricultural practices and user expectations (Nguyen et al., 2024; Ahmed et al., 2023).

Furthermore, RAD supports modular development, allowing different components of the DSS, such as the predictive model interface, database, and visualization layer, to be developed independently and later integrated efficiently. This modular approach improves system flexibility, scalability, and maintainability, which are essential for modern web-based agricultural systems (Wolfert et al., 2021).

In addition, the integration of cloud-based technologies further enhances DSS capabilities by enabling real-time data processing, remote access, and scalability. Cloud-enabled DSS allow users to access the system from different locations and ensure that predictions are based on up-to-date environmental data (Omar et al., 2021; Wolfert et al., 2021).

The literature emphasizes that Decision Support Systems help transform agricultural data into actionable insights for better decision-making. Modern DSS integrate predictive analytics, visualization tools, machine learning, and cloud-based technologies to improve crop planning and farm management. Studies also identified Rapid Application

Development (RAD) as an effective methodology because of its focus on rapid prototyping, modular design, and continuous user feedback. These findings support the development of a flexible and user-centered DSS for YieldShield. Effective DSS implementation can also improve the efficiency and accuracy of agricultural management practices.

System Usability Evaluation

System usability evaluation is a critical component in assessing the effectiveness of information systems, particularly those that involve user interaction with complex data, such as DSS. These systems often present analytical outputs, predictive results, and data visualizations that require users to interpret information accurately in order to make informed decisions. As a result, usability becomes a key factor in determining whether a system can effectively support its intended purpose. It is not sufficient for a system to be technically functional; it must also be accessible, intuitive, and efficient for its target users. In the context of agricultural systems, where users may have varying levels of technical expertise, usability plays an even more significant role in ensuring that predictive insights are properly understood and applied (Lewis & Sauro, 2021; Perez et al., 2022).

Usability is commonly defined based on established frameworks that emphasize three core dimensions: effectiveness, efficiency, and user satisfaction. Effectiveness refers to the accuracy and completeness with which users achieve specific goals using the system. Efficiency relates to the resources, such as time and effort, required to accomplish these tasks. User satisfaction, on the other hand, reflects the overall comfort and acceptability of the system from the user's perspective. These dimensions collectively highlight that usability is not solely determined by system performance, but also by the interaction

between the user, the system, and the context in which it is used (Brooke, 1996; Tan & Lim, 2021). In agricultural decision support systems, these factors are particularly important because users must be able to quickly interpret outputs such as crop yield predictions and recommendations without extensive technical training. Studies have shown that systems with high usability significantly improve user engagement and decision-making efficiency, especially in environments where timely decisions are critical (Ahmed et al., 2023).

To evaluate usability in a standardized and reliable manner, the System Usability Scale (SUS) has been widely adopted across various domains, including web-based platforms, mobile applications, and decision support systems. Developed by Brooke (1996), SUS is a simple yet effective tool consisting of a 10-item questionnaire that uses a five-point Likert scale to measure users' perceived usability of a system. Each item alternates between positive and negative statements, allowing for a balanced assessment of user perception. The responses are then converted into a single score ranging from 0 to 100, which represents the overall usability of the system. One of the key advantages of SUS is its ease of use, as it can be administered quickly without requiring extensive training or resources. Furthermore, SUS has been validated through numerous studies and is known for producing consistent and reliable results across different types of systems (Bangor et al., 2021; Lewis & Sauro, 2021).

Several studies have demonstrated the applicability and effectiveness of SUS in evaluating agricultural and information systems. For instance, Nguyen et al. (2024) utilized SUS to assess the usability of mobile-based agricultural platforms and found that it provides a reliable measure of user satisfaction and system performance. Similarly, Santos

and Dela Cruz (2025) highlighted that SUS is effective in determining whether a system meets acceptable usability standards, particularly in the context of farm management systems. These studies emphasize that SUS is not only practical but also adaptable to different types of systems, making it a widely accepted tool for usability evaluation in both academic and real-world applications.

Despite its advantages, SUS also has several limitations that must be considered when interpreting its results. One of its primary limitations is that it measures perceived usability rather than actual system performance. As a subjective evaluation tool, SUS reflects users' impressions and experiences, which may not always align with objective performance metrics such as task completion time, error rates, or system efficiency. For example, users may perceive a system as easy to use due to its visual design, even if underlying processes are inefficient (Lewis & Sauro, 2021). Additionally, SUS is considered a non-diagnostic tool, meaning that while it provides an overall usability score, it does not identify specific areas of the system that require improvement. This limitation makes it less suitable for detailed usability analysis unless combined with other evaluation methods (Tan & Lim, 2021).

Usability is particularly critical in decision support systems because these systems often present complex analytical outputs that require careful interpretation. Poor usability can increase cognitive load, making it difficult for users to process information and leading to potential errors in decision-making. It may also reduce system adoption, as users are less likely to engage with systems that are difficult to navigate or understand. On the other hand, well-designed systems enhance clarity, reduce complexity, and support efficient interaction, enabling users to translate data insights into actionable decisions more

effectively. Visualization tools, such as dashboards and graphical representations, play a key role in improving usability by simplifying complex data and making patterns more easily recognizable (Perez et al., 2022; Reyes et al., 2023).

Recent studies also emphasize the importance of integrating usability evaluation throughout the system development lifecycle rather than treating it as a final step. Continuous usability testing allows developers to identify issues early, refine system features, and improve user interaction over time. This iterative approach ensures that systems remain aligned with user needs and expectations, ultimately resulting in more effective and user-centered solutions (Ahmed et al., 2023; Nguyen et al., 2024). In agricultural contexts, this is especially important because users may have diverse backgrounds, varying levels of digital literacy, and different requirements depending on their specific farming practices.

Overall, the reviewed literature highlights that usability evaluation is important in ensuring that systems are efficient, user-friendly, and easy to understand. The System Usability Scale (SUS) was commonly identified as a reliable tool for measuring user satisfaction and perceived usability. Researchers also emphasized that continuous usability testing improves system functionality and user interaction. These findings support the integration of usability evaluation in YieldShield to ensure effective and accessible decision-making support for users. High system usability can also increase user acceptance and encourage continuous system utilization.

CHAPTER 3

TECHNICAL BACKGROUND

Project Design

The project adopts a developmental research design with embedded quantitative analysis. The developmental component focuses on the design, development, and evaluation of the proposed system, YieldShield: A Predictive Crop Yield Intelligence System, which aims to provide crop yield predictions and crop recommendations based on agricultural and environmental data. This approach directly supports the objective of designing and developing an integrated system that delivers predictive insights and decision-support tools through a user-friendly dashboard.

The embedded quantitative analysis component is utilized through the collection, processing, and analysis of numerical data such as environmental variables (e.g., rainfall, temperature, humidity), soil characteristics, land features, and historical crop yield records obtained from the DA-Binalonan. These datasets are prepared through systematic data cleaning, organization, and preprocessing techniques to ensure suitability for analysis. This directly aligns with the objective of collecting, processing, and organizing agricultural and environmental data for crop yield analysis.

Furthermore, this project develops and designs a predictive model using appropriate machine learning techniques and selects which among the applied techniques will provide better accuracy to estimate crop yield based on identified key factors. The performance of the model is evaluated using quantitative metrics such as RMSE, MAE, and R^2 to determine its reliability and effectiveness.

Quantitative analysis is further applied in the evaluation of the system's usability through the use of the SUS. The SUS enables the collection and analysis of numerical user feedback regarding the system's usability. By quantifying user experience through standardized scoring, the study ensures that the system's interface and functionality are assessed using measurable and statistically interpretable data. This supports data-driven improvements and validates the effectiveness of the system from the users' perspective.

The project utilizes a hybrid methodology combining the CRISP-DM Framework and Rapid Application Development (RAD) model.

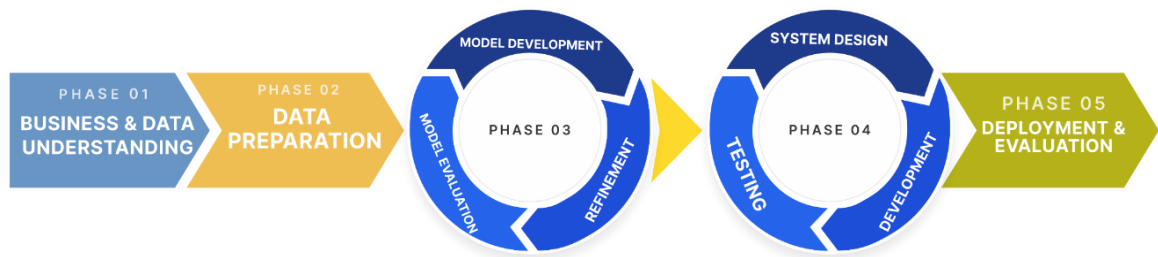


Figure 3. Hybrid CRISP-DM & RAD Framework

The CRISP-DM framework will be applied in handling data-related processes such as data collection, preparation, modeling, and evaluation, ensuring a structured approach to achieving accurate predictions. Meanwhile, the RAD model supports rapid system prototyping, iterative development, and user feedback integration, which is essential in developing a responsive and user-centered system. This combined approach ensures that both the analytical and system development objectives of the study are effectively achieved.

Overall, this research design ensures that the proposed system is both functionally effective and data-driven, supporting accurate predictions and user-centered decision-

making. The following phases of the combined CRISP-DM and RAD methodology are applied in the project:

Business and Data Understanding. This phase will establish the overall direction of the project by identifying the needs of farmers and agricultural stakeholders, as well as the objectives and scope of the YieldShield system. Initial agricultural and environmental datasets will be collected and examined to assess their relevance to crop yield prediction.

Key variables that influence crop yield such as rainfall, temperature, soil conditions, and crop type will be identified. This will ensure that the model will be developed using meaningful, relevant, and high-impact features.

Data Preparation. In this phase, the collected datasets will be prepared for model development. This will include selecting relevant records, handling missing values, and organizing the data into a consistent and usable format.

The objective will be to achieve a clean and workable dataset within a short timeframe, allowing model development to proceed without unnecessary delays caused by over-processing.

Model Development, Evaluation and Refinement. This phase focuses on developing the predictive model, which serves as the core component of the system. Multiple machine learning algorithms will be implemented and evaluated to analyze the relationships between environmental variables and crop yield. Candidate models may include techniques such as Random Forest and other suitable algorithms. The model with the best performance, based on evaluation metrics such as accuracy and error measures, will be selected as the final predictive model.

This phase focuses on developing the predictive model, which serves as the core component of the system. Multiple machine learning algorithms will be implemented and evaluated to analyze the relationships between environmental variables and crop yield. Candidate models may include techniques such as Random Forest and other suitable algorithms. The model with the best performance, based on evaluation metrics such as accuracy and error measures, will be selected as the final predictive model.

The model will be trained using historical agricultural data, where crop yield values will serve as the target variable. Through this process, the model will learn patterns and relationships within the data and will generate crop yield predictions based on input conditions.

The emphasis will be on producing a functional and reliable model capable of delivering accurate predictions, rather than on exhaustive parameter optimization at the initial stage.

Once the predictive model is developed, it will be evaluated using appropriate performance metrics, such as accuracy measures, error rates, and consistency indicators. The results will be analyzed to determine the model's predictive capability.

If necessary, refinements will be applied, including feature adjustments and model retraining, to improve performance. This phase will ensure that the model meets acceptable standards of accuracy and reliability prior to system integration.

System Design, Development, and Testing. Following model validation, the dashboard and decision support system (DSS) will be developed. This phase will focus on designing a user-friendly interface that will enable users to input data and interpret model outputs effectively.

The system will present crop yield forecasts, recommended crops, and visualizations such as trends and analytical insights. The dashboard will function as the primary interface that connects users with the predictive model.

The system will undergo iterative testing with target users, including farmers and agricultural personnel. Features will be introduced progressively and refined based on user feedback.

System usability will be evaluated using the System Usability Scale (SUS), which will assess factors such as ease of use, efficiency, and user satisfaction. The results will guide improvements in system design and functionality.

Deployment and Evaluation. In the final phase, the fully developed system will be deployed in a real-world environment. The predictive model and dashboard will be integrated into a complete and operational system accessible to end users.

Comprehensive testing will be conducted to ensure system stability, performance, and reliability. Users will be oriented on how to effectively utilize the system for crop yield prediction and decision-making. Continuous monitoring will be carried out to maintain system performance and address any emerging issues.

The project adopts a Logic Model approach to describe the overall operation of the system as depicted in the next figure.



Figure 4. Conceptual Framework of the Project

The conceptual framework of the study is based on the Logic Model approach, which illustrates the systematic transformation of inputs into outputs and outcomes through a series of structured processes. This framework provides a clear representation of how the proposed system, YieldShield: A Predictive Crop Yield Intelligence System, utilizes agricultural and environmental data to generate meaningful insights that support data-driven decision-making in agriculture.

The input component represents the resources and data required for the development and operation of the system. These inputs include historical agricultural datasets such as crop yield records, soil characteristics, and land features, as well as environmental data including rainfall, temperature, humidity, and other climate-related variables. In addition, real-time weather data obtained through external APIs are incorporated to ensure that predictions are based on current environmental conditions. User-provided farm data, such as location and crop type, are also included to enable personalized analysis. Furthermore, system requirements and technical tools, including programming languages, machine learning libraries, and database systems, serve as foundational resources necessary for system development and implementation.

The activities component represents the processes involved in transforming the inputs into meaningful outputs. This begins with data collection and preprocessing, where raw datasets are cleaned, organized, and prepared to ensure quality and consistency. Feature engineering is then applied to extract relevant variables and derive meaningful indicators that enhance model performance. The processed data are used to develop a predictive model using applicable Machine Learning Algorithms, wherein the algorithm that yields the highest accuracy will be used for the predictive model of the proposed

system. Model training and evaluation are conducted using appropriate performance metrics to ensure accuracy and reliability. In parallel, the system is developed using a hybrid methodology combining CRISP-DM and Rapid Application Development (RAD), enabling efficient data handling, iterative prototyping, and user-centered design. Finally, the predictive model is integrated into a web-based dashboard, allowing users to interact with the system and visualize results.

The output component represents the immediate results generated by the system. These outputs include crop yield predictions, which provide quantitative estimates of expected agricultural production based on input conditions. The system also generates crop recommendations by identifying the most suitable crops for specific environmental and land characteristics. Additionally, an interactive dashboard presents visualizations such as trends and insights, enabling users to easily interpret the results. The system further identifies key factors influencing crop yield through feature importance analysis and produces reports that support planning and decision-making processes.

The outcomes component reflects the long-term impact and benefits of the system. By providing accurate and timely predictions, the system enhances agricultural decision-making, allowing farmers and stakeholders to make informed choices regarding crop planning and resource allocation. This leads to improved productivity and optimized use of resources such as land, water, and inputs. Moreover, the system helps reduce risks associated with climate variability by enabling proactive planning. Ultimately, YieldShield promotes sustainable and resilient farming practices by integrating data-driven technologies into agricultural processes and making advanced predictive insights accessible to users.

The use of the Logic Model framework is appropriate for this study as it clearly demonstrates the relationship between data inputs, analytical processes, system outputs, and long-term agricultural outcomes. It provides a structured overview of how raw agricultural data are transformed into actionable insights, supporting the overall objective of improving crop yield prediction and agricultural decision-making.

Requirements Analysis

The Requirements Analysis will focus on identifying, examining, and validating the needs of farmers, agricultural officers, and system administrators, as well as the technical requirements necessary for the development of the YieldShield system. This process will ensure that the system is grounded on actual agricultural practices and will effectively address challenges related to crop planning, yield prediction, and data-driven decision-making in the agricultural sector.

System requirements will be gathered using structured methods to ensure completeness, accuracy, and alignment with the objectives of the study.

Unstructured Interviews. Unstructured interviews will be conducted with agricultural officers from the Department of Agriculture Office of Binalonan, Pangasinan, as well as selected farmer representatives from each of the 24 barangays in the municipality. The participants will include approximately six agricultural officers and 24 farmer representatives (one per barangay), serving as potential users of the YieldShield system. These interviews aim to gather insights into current agricultural practices, challenges in crop planning, and decision-making processes.

The discussions will focus on how farmers access agricultural information, how they assess environmental conditions, and the difficulties they encounter in predicting crop yield. The information collected will support Objective 1, by identifying the relevant agricultural and environmental data needed for crop yield analysis, and Objective 4, by informing the design of the system workflow and dashboard interface. These interviews will help identify user needs, system expectations, and opportunities where YieldShield can improve accessibility, efficiency, and data-driven decision-making.

Document Analysis. Relevant agricultural datasets, reports, and records obtained from the Department of Agriculture – Binalonan and other reliable sources will be reviewed to determine the key variables required for crop yield prediction. These variables include environmental factors such as rainfall, temperature, humidity, and weather conditions, as well as agricultural data such as soil characteristics, terrain, crop types, land area, and historical yield records. This analysis supports Objective 1 by ensuring the collected data are accurate, relevant, and suitable for model development, and informs Objective 2 by identifying the most significant predictors to include in the crop yield prediction models. The document analysis ensures that the predictive models are built on reliable datasets and reflect real-world agricultural conditions.

Workflow Analysis. Existing agricultural decision-making processes will be examined through workflow analysis and process mapping. This involves identifying how farmers currently plan crop selection, monitor environmental conditions, and make yield-related decisions. Step-by-step activities, including data gathering, observation, consultation with agricultural officers, and reliance on experience-based practices, will be documented. This analysis addresses Objective 4, as it informs the design of an efficient

system workflow and user interface for the YieldShield dashboard, highlighting areas where interactive heatmaps and notifications can improve decision-making, reduce inefficiencies, and support timely agricultural planning.

System Usability Testing. To evaluate the usability of the proposed YieldShield system, a system evaluation will be conducted using the System Usability Scale (SUS). Thirty participants, consisting of six DA officers and 24 farmer representatives (one from each barangay), will be selected through purposive sampling. This method directly supports Objective 5, as it gathers feedback from individuals who will use the system, measuring effectiveness, efficiency, and user satisfaction. Insights from the SUS evaluation will help refine the dashboard interface, notifications, and heatmaps, ensuring the system meets user needs and provides a practical, user-friendly tool for data-driven agricultural decision-making.

Respondents will interact with the YieldShield system and answer a standardized 10-item questionnaire, with each item rated using a 5-point Likert scale.

To determine the system's total usability score, the responses will be computed using the standard SUS scoring procedure and converted into a 0–100 scale. The SUS score for each respondent will be calculated using the following formula:

$$SUS = \left(\sum_{i=1,3,5,7,9} (X_i - 1) + \sum_{i=2,4,6,8,10} (5 - X_i) \right) \times 2.5$$

where:

- X_i = score given by the respondent for each item
- Odd-numbered items (1, 3, 5, 7, 9) are scored as $(X_i - 1)$
- Even-numbered items (2, 4, 6, 8, 10) are scored as $(5 - X_i)$

The resulting scores will be interpreted using the SUS adjective rating scale and acceptability ranges, as shown in Table 1.

Table 1. SUS Score Interpretation Scale

SUS Score Range	Adjective Rating	Acceptability
80.3 – 100	Excellent	Acceptable
68 – 80.2	Good	Acceptable
68	OK	Marginal
51 – 67	Poor	Marginal
0 – 50	Worst	Not Acceptable

The system's usability will be interpreted based on these scores, where a result of 68 or higher indicates that the application has reached an acceptable level of usability. Conversely, scores below this threshold suggest that the system may present usability issues and require further improvement. This approach ensures that the evaluation reflects the perspectives of actual end-users in the agricultural sector and supports the identification of system improvements aligned with their needs.

The gathered requirements will be systematically organized and analyzed to ensure that they are complete, consistent, and aligned with the objectives of the system. This will ensure that the final system design will support effective data management, accurate predictive modeling, reliable system integration, and user-friendly interaction for agricultural stakeholders.

To support system modeling and visualization, the following tools will be utilized:

Use Case Diagram. The Use Case Diagram will define the interactions between the system users farmers and administrators and the YieldShield system. It will illustrate key functionalities such as inputting agricultural and environmental data, generating crop yield predictions, receiving crop recommendations, viewing dashboard visualizations, and generating reports. This directly supports Objective 4, as it ensures that all required system

features are clearly identified, aligned with user needs, and incorporated into the design of the system workflow and dashboard interface. By mapping these interactions, the diagram provides a clear blueprint for system development, ensuring that the YieldShield platform meets its intended goals of improving accessibility, efficiency, and data-driven decision-making in agricultural planning.

System Flowchart. The System Flowchart will illustrate the internal processing logic of the YieldShield system, showing how raw agricultural and environmental data are collected, preprocessed, and feature-engineered before being processed by the Random Forest predictive model. The model will generate crop yield forecasts and suitable crop recommendations based on the given conditions, using historical datasets as training data where crop yield values serve as the target variable. The flowchart also demonstrates how the outputs are presented through the system dashboard and reports, enabling users to interpret results and make informed decisions. This directly supports Objective 4, as it guides the development of an efficient system workflow and user interface, ensuring that YieldShield provides actionable insights and facilitates data-driven decision-making for both farmers and administrators.

All data handling procedures will comply with applicable data privacy and ethical standards. Agricultural data will be obtained only from authorized and reliable sources, and no personally identifiable information will be collected or processed. The system will focus solely on environmental and agricultural variables necessary for analysis. Data will be used strictly for academic and system development purposes. Access to datasets will be controlled, and appropriate measures will be implemented to ensure data integrity, confidentiality, and security throughout the development process.

System Requirements

Hardware

The table below presents the hardware requirements necessary for the development and operation of the YieldShield system. These specifications outline the minimum and recommended computing resources needed to ensure efficient system performance, data processing, and user interaction.

Table 2. Development & Deployment Hardware Specifications

Environment	Component	Minimum Specification	Recommended Specification
Development	Processor	Dual-core (Intel i3 / Ryzen 3 equivalent)	Quad-core (Intel i5 / Ryzen 5 or higher)
	RAM	8GB	16 GB
	Storage	5–10 GB free space	SSD with 20 GB+ free space
	Operating System	Windows 10	Windows 11
Deployment (Client)	Device Type	Smartphone / Basic Laptop	Modern Smartphone / Laptop / Desktop
	RAM	2GB	4GB or Higher
	Internet Connection	3G / basic mobile data	4G/5G or stable broadband
Cloud Server	Server Type	Cloud Machine	Virtual Cloud Virtual Machine
	Processor	2 vCPU	4 vCPU
	RAM	4 GB	8 GB
	Storage	50 gb	80 gb

The hardware requirements are structured to support the distinct demands of system development, deployment, and end-user access. For development, a multi-core processor and higher memory capacity are required to handle simultaneous execution of the web framework, database services, and R-based analytical processes. These tasks involve data

preprocessing, model training, and interface development, which demand stable performance and sufficient computing resources.

For deployment, the system is designed to operate on moderate server specifications to balance performance and cost-efficiency. The defined processor, memory, and storage capacity are sufficient to manage database transactions, process user requests, and run the predictive model without significant latency under typical usage conditions.

On the client side, lower minimum specifications are intentionally set to accommodate users with basic smartphones or computers. This ensures that the system remains accessible to agricultural stakeholders while still providing acceptable performance. Since the proposed system is web-based, it primarily relies on a stable internet connection and sufficient RAM to support basic browser operations.

To support the deployment environment of the system, a cloud server configuration is also defined. This ensures that YieldShield can be hosted in an online environment where data storage, processing, and user access are centrally managed and consistently available. The cloud server is responsible for handling incoming user requests, managing the database, and executing the predictive analytics component of the system in real time. It also provides scalability and reliability, allowing the system to operate continuously without being dependent on a single physical machine. In line with the system's computational requirements and expected usage load, a moderate cloud infrastructure is specified. This configuration is sufficient to support web application hosting, database

operations, and lightweight machine learning inference while maintaining system responsiveness and cost efficiency.

Software

Table 3 outlines the software requirements needed to develop, deploy, and maintain the YieldShield system. It includes the essential tools, platforms, and technologies that support system functionality, data processing, and interface development.

Table 3. Software Requirements

Environment	Software	Description
Development	Programming Language	Node.js, React.js, R programming Language
	Database	Postgresql
	Development Tools	VS Code, RStudio
	Browser	Google Chrome, Microsoft Edge, Firefox, Chromium-based
Deployment (Server)	Operating System	Windows
	Web Server	Apache
	Database Server	Postgresql
	Browser	Any modern web browser
	Internet Connection	Required
(Client)		

The selected software components are chosen to support both system development and deployment efficiently. Node.js and React.js are used to build a responsive and scalable web-based interface, while the R programming language enables statistical analysis and predictive modeling for crop yield estimation. PostgreSQL is utilized as a reliable database management system capable of handling structured agricultural data. Development tools such as VS Code and RStudio facilitate efficient coding, debugging, and model development. For deployment, a Linux-based environment with an Apache web server is adopted due to its stability, security, and widespread use in web applications. The inclusion of modern web browsers ensures that the system remains accessible across different devices without requiring additional software installation.

Peopleware

Table 4 identifies the peopleware requirements of the system, specifying the individuals involved in its development, management, and usage. It highlights the roles and responsibilities necessary to ensure the effective implementation and utilization of the YieldShield system.

Table 4. Roles and Responsibilities

Role	Responsibility
The Proponents	Design, develop, and implement the YieldShield system, including the predictive model and user interface.
System Administrator	Manage system deployment, database maintenance, and ensure system security and performance.
Agricultural Officers	Provide agricultural data, assist in validating system outputs, and support farmers in using the system.
Farmers (End Users)	Input relevant farm data and utilize the system's predictions for decision-making in crop planning.
Researchers/Proponents	Analyze data, evaluate system performance, and ensure the accuracy and reliability of the predictive model.

These stakeholders include the proponents or researchers, system administrator, agricultural officers, and farmers, all of whom contribute to the system's overall success.

The proponents are responsible for designing and developing the system, including its predictive model and user interface, as well as evaluating its performance and ensuring its reliability. The system administrator handles deployment, maintenance, database management, and system security to ensure stable and continuous operation. Agricultural officers serve as domain experts who validate system outputs and ensure that recommendations align with actual farming practices, while farmers act as the primary end users by providing field data and utilizing the system's predictions for informed decision-making in crop planning. The collaboration of these stakeholders ensures that the system is functional, accurate, and applicable to real-world agricultural needs.

Network

The proposed system requires a stable and reliable network infrastructure to ensure continuous access and efficient data processing. An internet connection is essential, as it enables end-users such as agricultural officers and farmers to access the proposed system through their devices and submit or retrieve agricultural data in real time. During development and testing, a Local Area Network (LAN) is utilized to allow controlled communication between development machines and local servers within a limited environment.

For deployment, the system relies on the public internet, which serves as the communication medium connecting client devices to the hosted server. This setup allows users from different geographical locations to access the system remotely without being restricted to a single physical network. The system will be hosted on a cloud-based platform, which will handle data storage, processing of agricultural datasets, and execution of predictive models.

To maintain system performance, a minimum bandwidth of 5 Mbps is required for basic access and data submission, while 10–25 Mbps is recommended to support multiple users, real-time data updates, and smoother interaction with the system. This ensures that data transmission and predictive processing occur without noticeable delays.

In terms of security, the system implements standard protection mechanisms including user authentication (login credentials), role-based access control, and secure data transmission using HTTPS (SSL/TLS encryption). These measures help protect sensitive agricultural data from unauthorized access and ensure that communication between clients and the server remains secure.

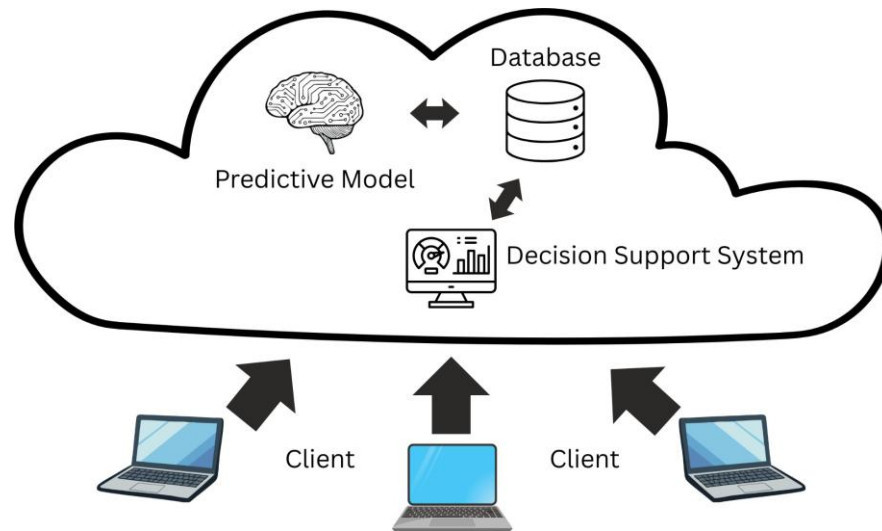


Figure 5. Logical Architecture Diagram

The proposed system follows a client-server model within a cloud-based environment. Multiple client devices connect to the system via the Internet, allowing users to input agricultural data and access results. Within the cloud, the core components include the predictive model, database, and decision support system. The predictive model processes input data to generate crop yield predictions, while the database stores both raw and processed data. The decision support system then presents the results in a user-friendly format. These components continuously interact to ensure efficient data flow and real-time processing. This setup supports scalability, reliable performance, and effective delivery of predictive insights for agricultural decision-making.

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APPENDIX A

SUS QUESTIONNAIRE

Name (Optional): _____

Age: _____ Sex: _____

A. Instructions

Please indicate your level of agreement with the following statements based on your experience using the **YieldShield system** by placing a check in the appropriate box.

(Panuto: Pakisagot ang mga sumusunod na pahayag batay sa inyong karanasan sa paggamit ng sistemang YieldShield sa pamamagitan ng paglalagay ng tsek sa tamang kahon.)

Rating Scale:

5 – Strongly Agree (Lubos na sumasang-ayon)

4 – Somewhat Agree (Medyo sumasang-ayon)

3 – Neutral (Walang opinyon)

2 – Somewhat Disagree (Medyo hindi sumasang-ayon)

1 – Strongly Disagree (Lubos na hindi sumasang-ayon)

Statements (Pahayag)	Strongly Disagree 1	Somewhat Disagree 2	Neutral 3	Somewhat Agree 4	Strongly Agree 5
1. I think i would like to use this tool frequently. (sa tingin ko ay nais kong gamitin ang sistemang ito nang madalas.)					
2. I found the tool unnecessarily complex. (nalaman ko na ang sistema ay hindi kailangang maging kumplikado.)					
3. I thought the tool was easy to use. (naisip ko na ang sistema ay madaling gamitin.)					

4. I think that i would need the support of a technical person to be able to use this system. (sa tingin ko ay kailangan ko ng tulong ng teknikal na tao upang magamit ang sistemang ito.)					
5. I found the various functions in this tool were well integrated. (maayos ang pagkaka-integrate ng mga function ng sistema.)					
6. I thought there was too much inconsistency in this tool. (masyadong hindi pare-pareho ang sistema.)					
7. I would imagine that most people would learn to use this tool very quickly. (madaling matutunan gamitin ng karamihan ang sistema.)					
8. I found the tool very cumbersome to use. (mahirap at hindi komportable gamitin ang sistema.)					
9. I felt very confident using the tool. (kumpiyansa ako sa paggamit ng sistema.)					

10. I needed to learn a lot of things before i could get going with this tool. (kailangan ko pang matutunan ang maraming bagay bago magamit ang sistema.)					
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